

The Worldsheet and the Copper Plane

A string scattering amplitude is the thermal-noise statistics of a resistive sheet

Model an interacting particle not as a zero-dimensional point but as an extended one-dimensional string, sweeping out a continuous two-dimensional worldsheet in spacetime. Its quantum probability amplitude is then, integrand for integrand, the same mathematical object as the statistics of thermal fluctuations of electrical power across a two-dimensional resistive copper plane. In the correspondence the momenta of the incoming and outgoing particles become discrete electrical currents, injected and extracted at coordinate locations z_i strictly along the one-dimensional outer boundary of the board. A

two-dimensional sheet has a logarithmic Green's function, so the power dissipated by any Kirchhoff-balanced set of edge currents is a sum of pairwise logarithms of the contact separations; and Johnson–Nyquist noise weights each microscopic thermal state by the exponential of its dissipated heat. Integrating those thermal states by sliding the injection points continuously along the boundary therefore reproduces the quantum path integral of the interacting strings, governed by the mathematical equivalence:

$$\int e^{-P} dz_1 \cdots dz_N = \int \prod_{1 \leq i < j \leq N} |z_i - z_j|^{2\alpha' p_i p_j} dz_1 \cdots dz_N$$

READ AS A COPPER PLANE

the Boltzmann weight of the dissipated power

READ AS A STRING WORLDSHEET

the Koba–Nielsen factor of the N -point amplitude

NOT A NEW OBSERVATION — THE CHAIN OF RESULTS IT RESTS ON

1730 Euler

the integral that interpolates the factorial, to Goldbach, 8 January.

1928 Johnson · Nyquist

measure, then explain, resistor noise. Phys. Rev. 32, 97 & 110.

1968 Veneziano

the crossing-symmetric four-point amplitude. Nuovo Cimento A57, 190.

1969 Koba · Nielsen

generalise it to N particles — the product above. Nucl. Phys. B10, 633.

1970 Fairlie · Nielsen

the analogue model: the amplitude as a flow's dissipation. B20, 637.

1981 Polyakov

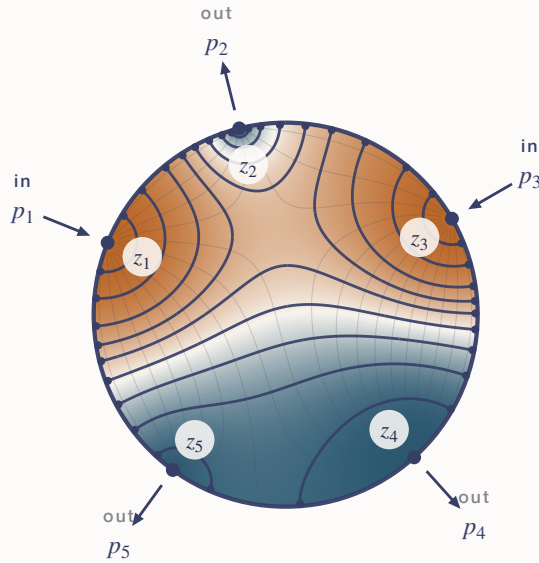
makes the worldsheet a 2-D statistical system. Phys. Lett. B103, 207.

One integrand, two readings

The same harmonic field on the same disk, with the same five boundary points

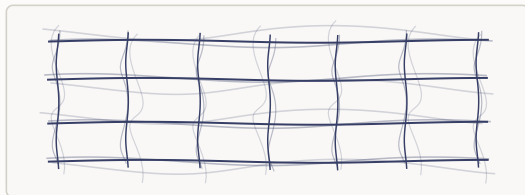
The string worldsheet

a 1-D string sweeping a 2-D surface



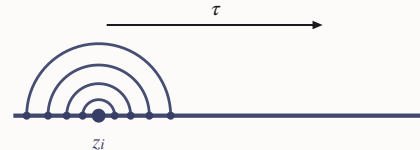
$$\langle X^\mu(z_i) X^\nu(z_j) \rangle = -2\alpha' \eta^{\mu\nu} \ln |z_i - z_j|$$

the worldsheet field, in spacetime



Microscopic fluctuations: quantum

The embedding X^μ fluctuates; each history weighted $\exp(-S)$.



a bold curve is the string at one instant; its ends are free on the rim, meeting it at 90° ; and near a puncture it shrinks on the page, not in fact

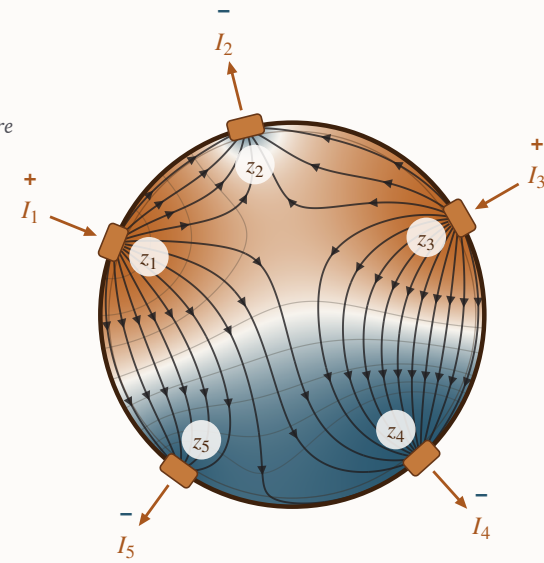


ONE INTEGRAND TWO READINGS

$$\begin{aligned} p_i &\leftrightarrow I_i \\ X^\mu(z) &\leftrightarrow V(z) \\ \alpha' &\leftrightarrow R_s / \pi P_N \\ e^{-S} &\leftrightarrow e^{-P/P_N} \\ \Sigma p_i = 0 &\leftrightarrow \Sigma I_i = 0 \end{aligned}$$

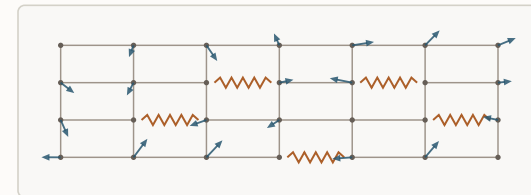
The resistive copper plane

a 2-D sheet dissipating heat



$$V(w) = -(R_s / \pi) \sum_i I_i \ln |w - \zeta_i|$$

the potential — exact, because $\Sigma I = 0$



Microscopic fluctuations: thermal

Every element rattles; each pattern weighted $\exp(-P/P_N)$.

Colour is one scalar field, drawn identically on both sides — copper positive, teal negative; the flow lines are the exact current density $J = -\nabla V / R_s$.

Read the currents as light-cone momenta and Mandelstam's map gives $\tau = -(\pi / R_s) V$: voltage is the string's time, so a level curve IS the string at an instant.

Five steps, each one measured

Every ✓ below is a number produced by `verify_string_copper.py`, not a citation

THE CHAIN, IN ORDER

1 The string side is a product of powers.

An open string's vertex operators sit on the one-dimensional boundary of the worldsheet, where the embedding has the propagator $\langle X^\mu(x)X^\nu(y) \rangle = -2\alpha' \eta^{\mu\nu} \ln|x-y|$. Contracting N tachyon operators $e^{ip \cdot X}$ turns that logarithm into the Koba–Nielsen factor $\prod |z_i - z_j|^{2\alpha' p_i \cdot p_j}$.

✓ the product is exp of a logarithmic pair energy, to 4.4×10^{-16}

2 A sheet's rim Green's function is $-R_s/\pi$ times a log.

A two-dimensional sheet has a logarithmic Green's function, and a contact ON an insulating rim sees twice the interior value, because its image charge sits on top of it. That factor of two is not bookkeeping: it is the same two that separates the open-string boundary propagator from the closed-string bulk one.

✓ measured -0.317971 against $-R_s/\pi = -0.318310$, 0.11%, converging as the sheet grows

3 Neutral rim currents dissipate a sum of logs.

Feeding that into $P = \sum_i I_i V(z_i)$ gives $P = -(2R_s/\pi) \sum_{i < j} I_i I_j \ln|z_i - z_j| + \kappa \sum_i I_i^2$. There is no factor of one half here — that belongs to the energy of assembling static charges, not to the power delivered by current sources, and inserting it halves α' .

✓ fitted 0.637208 against $2R_s/\pi = 0.636620$ over 300 random neutral patterns, $R^2 = 0.9999992$

AND ITS TWO CLOSING MOVES

4 Nyquist supplies the exponential — and its scale.

Thermal noise makes the contact currents a Gaussian whose quadratic form is exactly the dissipated power, so a fluctuation's weight is $\exp(-P/P_N)$ with $P_N = 4k_B T/\tau$ for an averaging window τ . Writing $\exp(-P)$ by itself is dimensionally incomplete: a power is not an exponent. Substituting step 3 then gives the Koba–Nielsen factor outright, with $\alpha' = R_s \tau / 4\pi k_B T$ — the Regge slope as a sheet resistance in units of noise power, with the right units of one over current squared.

✓ Ohmic flow strictly minimises P : a divergence-free loop of strength ϵ costs exactly $4\epsilon^2$

5 Gauge-fix three contacts, then slide the rest.

Fixing $z_1, z_3, z_4 = 0, 1, \infty$ and integrating the last over $(0, 1)$ collapses the $N = 4$ product to $\int x^{-\alpha(s)-1} (1-x)^{-\alpha(t)-1} dx = B(-\alpha(s), -\alpha(t))$, the Veneziano amplitude, with $\alpha(s) = 1 + \alpha's$. Euler's factorial-interpolation integral is the string amplitude.

✓ the integral equals Euler's B to 4.4×10^{-16}

THREE CONSEQUENCES THAT ARE PRECISE RATHER THAN POETIC

The three orderings are the three channels.

Four contacts admit three distinct cyclic orders around the rim. Those are the three integration ranges of x , and they give $B(-\alpha(s), -\alpha(t))$, $B(-\alpha(t), -\alpha(u))$ and $B(-\alpha(s), -\alpha(u))$ — the s-t, t-u and s-u channels, whose sum is the crossing-symmetric total.

✓ all three verified to 1.9×10^{-10} or better

“Counting continuously” is an identity.

$B(k+1, n-k+1) = 1/[(n+1)C(n, k)]$ exactly, so the Beta function really does interpolate the binomial coefficient. The integral counts arrangements the way a binomial does, with a continuous index rather than a discrete one.

✓ 38 integer pairs, 0 mismatches

The particle spectrum is two pads touching.

The s-channel poles come from the endpoint $x \rightarrow 0$, where two contacts merge and the dissipated power diverges. Convergence needs $2\alpha' p_1 \cdot p_2 > -1$; the poles sit at $\alpha's = -1, 0, 1, 2, \dots$, which is the open-string spectrum $m^2 = (J-1)/\alpha'$.

✓ $B(a, b) \rightarrow 1/a$ confirmed as $a \rightarrow 0$

Six things the one-line version hides

None of these breaks the identity; each one bounds what it may be read to mean

One plane carries one spacetime dimension.

$p_i p_j$ is a Minkowski inner product, so the matrix of exponents is indefinite — for four on-shell tachyons its eigenvalues are $-3, 0, 5, 6$. A real scalar current can only produce $I_i I_j$, which is rank one and positive semi-definite. One copper plane therefore realises a single target dimension; D dimensions need D superposed planes, the timelike one carrying imaginary current. Reading a current as a momentum vector is the one step in the statement that cannot be taken literally.

The integral as written is still divergent.

The measure is exactly $SL(2, \mathbb{R})$ -invariant — measured ratio 1.000000000000 under four Möbius maps — and it is invariant precisely because $\alpha' p^2 = 1$; pushed off the mass shell the ratio becomes 0.0652. So $\int dz_1 \dots dz_N$ contains the infinite volume of the gauge group. Three punctures must be held fixed and only $N-3$ integrated, which makes the equation an identity between integrands first and between amplitudes only after.

The self-energy and the mass shell are one thing.

The $i=j$ terms diverge. On the sheet that is cut off by the finite radius of a real pad; on the worldsheet it is removed by normal ordering, and what normal ordering leaves behind is $\alpha' p^2 = 1$. Regulator and mass shell are the same object under two names.

Rim versus interior is open versus closed.

The factor of two between a contact on the insulating rim and one in the interior — measured as 1.99689 — is the same factor of two between the open-string boundary propagator $-2\alpha' \ln|x-y|$ and the closed-string bulk one $-\alpha' \ln|z-w|$. Moving a contact off the edge into the copper is not a small perturbation of the model: it converts an open-string insertion into a closed-string one.

This is the worldsheet CFT, and not a least-resistance path.

The two-dimensional theory here is the string's own worldsheet, and the copper plane is that worldsheet — not spacetime. It is not the boundary CFT of AdS/CFT, where the dual theory lives on the boundary of the spacetime the gravity fills; conflating those is the most inviting misreading of the statement. Nor is the amplitude a least-resistance path: Rayleigh's principle picks out only the most probable configuration, whereas the amplitude sums over all of them, which is the whole content of the exponential.

The two readings use different charges.

The strings on page 2 come from Mandelstam's map, which reads each current as a light-cone momentum p^+ — one component. The Koba–Nielsen weight reads the same current as the whole momentum vector. Two identifications, not one.

WHAT WAS MEASURED

independent of any citation — the sheet is a lattice of unit resistors, solved exactly

QUANTITY	MEASURED	EXACT	AGREEMENT
Rim-contact pair potential, $R_s = 1$	−0.317971	$-R_s/\pi = -0.318310$	0.11%
Interior-contact pair potential	−0.159233	$-R_s/2\pi = -0.159155$	0.05%
Rim / interior ratio	1.99689	2 exactly	the image charge
Power-law coefficient, 300 patterns	0.637208	$2R_s/\pi = 0.636620$	0.09%, $R^2 = 0.99999992$
Non-neutral ΔP per size doubling	0.8825	$(Q^2 R_s/\pi) \ln 2 = 0.8825$	0.01%
Gauge-fixed integral vs Euler B	0.26666666666667	$B(2, 1.5) = 0.26666666666667$	4.4×10^{-16}
$SL(2, \mathbb{R})$ measure ratio, on the mass shell	1.000000000000	1 exactly	four Möbius maps
the same, pushed off the mass shell	0.065232	$\neq 1$	the symmetry is the shell
Koba–Nielsen exponent eigenvalues	−3, 0, 5, 6	indefinite, rank 3	a real current gives rank 1, ≥ 0
Washer resistance, $a = 120, b = 600$	0.256748	$(1/2\pi) \ln(b/a) = 0.256150$	0.23%, first order in h

Punch a hole in the copper

Closed strings carry the graviton, and closed means the interior, not the rim

Gravity is a closed-string state.

	A CONTACT ON THE RIM	ONE IN THE INTERIOR
the insertion sits	on the 1-D rim	in the 2-D interior
so the string is	open	closed
its propagator	$-2\alpha' \ln x-y $	$-\alpha' \ln z-w $
lightest massless state	spin 1 — a gauge boson	spin 2 — the graviton

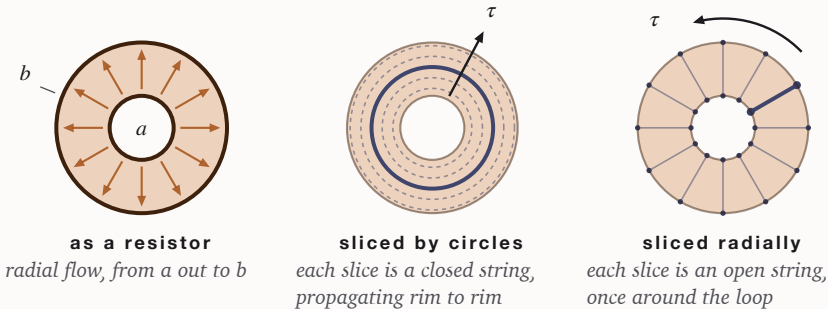
The factor of two I measured between those two propagators — 1.99689 — is therefore not a curiosity of lattice Green's functions. It is the gauge-theory/gravity divide. Open strings give the gauge fields that live on a boundary; closed strings give gravity in the bulk. Sliding a current contact off the board edge and into the copper crosses exactly that line.

But injecting a current is not a graviton.

The insertion this document analyses is $e^{ip \cdot X}$: a point charge, a monopole current source — and the state it creates is the tachyon, not the graviton. A graviton vertex operator is $\epsilon_{\mu\nu} \partial X^\mu \partial X^\nu e^{ip \cdot X}$ — one derivative holomorphic, one antiholomorphic — which is BILINEAR in the worldsheet field. Since X plays the part of the potential, ∂X is the current density — so a graviton insertion injects no net current at all. It couples to something quadratic in the local current, which in copper is the Joule dissipation density $R_s |J|^2$. That is the same structure as the worldsheet stress tensor, and it is the honest version of “the graviton couples to stress-energy”.

A washer is the one-loop diagram.

Copper with a hole has two rims, and that shape is the annulus — the one-loop open-string diagram. It can be sliced two ways, neither more correct:



What a “modulus” is.

A conformal map may stretch and bend a shape, but never shear it. Any washer maps onto any other with the same ratio b/a — so a washer has exactly one shape parameter that conformal maps cannot remove. That leftover number is called its modulus, and an amplitude must integrate over it, because it sums over every shape the surface can take. Here the modulus is $t = (1/2\pi) \ln(b/a)$; on a lattice of unit resistors $a = 120$, $b = 600$ give $R = 0.256748$ against $t = 0.256150$ — 0.23%.

Conformally the cylinder's length is $\ln(b/a) = 2\pi R/R_s$; resistance is separation. So a HIGH-resistance washer is the long-cylinder limit, where only the lightest closed states survive — graviton exchange between the rims; as $b/a \rightarrow 1$ the open-string loop takes over instead.

READING THIS PAGE CORRECTLY

EXACT — an identity

The integrand identity itself; the rim-versus-interior factor of two; and R/R_s as the annulus modulus. Theorems about one body of mathematics — and measured here, not cited.

ANALOGY — a dictionary

A current is one number, a momentum a spacetime vector: one plane, one dimension (p. 4). The copper is the WORLDSHEET — this graviton is a state on an auxiliary surface, not curvature of the room.

NOT CLAIMED

That a board gravitates beyond the pull of its mass; that measuring copper could test string theory; or that this is holography. A resistor network is not a quantum theory with a Regge slope.